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B.Tech. Degree VI Semester Regular/Supplementary Examination June 2023

ME 19-205-0607 ENERGY CONSERVATION AND ENVIRONMENT PROTECTION (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Understand the overview of World Energy Scenario and energy economics.
 CO2: Study the importance of energy management and energy auditing.
 CO3: Learn about the renewable energy sources.
 CO4: Understand the environmental impacts of energy use.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create
 PO – Programme Outcome

PART A(Answer **ALL** questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) What are the benefits of an energy balance analysis?	3	3	L2	2	7
(b) What are the drawbacks of a Simple Pay Back Period method?	3	3	L2	1	7
(c) What do you mean by flash steam? How it's being used as an energy efficiency improvement tool?	3	3	L2	2	7
(d) How does an oversized pump affect the energy efficiency?	3	3	L2, L3	2	7
(e) Write the concept of Pinch in Heat Exchanger.	3	3	L2	1	7
(f) Name and briefly discuss the different renewable energy sources.	3	3	L1	3	1,7,12
(g) List the various steps to reduce particulates and water pollution.	3	3	L2	4	7,10
(h) Explain how CO ₂ emission causes global warming.	3	3	L2	4	7,10

PART B

(4 × 12 = 48)

II. (a) Analyze the challenges and opportunities in transitioning to a sustainable and renewable energy system in the context of the current world energy scenario.	6	6	L4	1	7
(b) Assess the potential of renewable energy adoption in India and analyze the challenges and strategies for achieving a low carbon energy future in the country.	6	6	L4	1	7

OR

III. A company is considering investing in a new machine that will cost ₹10 lakh. The machine is expected to generate annual cash flows of ₹2 lakh for the next 5 years. The company's discount rate is 10%.	12	12	L5	2	7
(i) What is the NPV of this investment?					
(ii) Find the approximate IRR for this investment.					
(iii) State the acceptance/rejection status for NPV and IRR models.					



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		Marks	BL	CO	PO																				
IV.	(a) An energy audit firm is hired to spot the inefficiencies in the boiler and steam distribution system of their client industry. Describe potential opportunities for saving energy in this context.	8	L3	2	7																				
	(b) What are the different methods to recover waste heat?	4	L1	2	7																				
OR																									
V.	(a) Discuss the potential energy saving opportunities in refrigeration and air conditioning systems.	6	L3	2	7																				
	(b) Explain how past plant data analysis helps energy management.	6	L1	2	7																				
VI.	(a) Define cogeneration. Explain with the help of neat sketch, classification and types of cogeneration.	6	L2	2	7																				
	(b) Explain how heat recovery can be calculated with the help of composite curves. Explain with neat figures.	6	L2	1	7																				
OR																									
VII.	Construct Heat Exchanger Network with the help of Pinch Analysis for the data given below:	12	L4	1	7																				
<table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"><tr><th>Streams</th><th>Specific Heat, Cp</th><th>Source Temperature Ts (°C)</th><th>Target Temperature Tt (°C)</th></tr><tr><td>Hot Stream 1</td><td>2</td><td>150</td><td>60</td></tr><tr><td>Hot Stream 2</td><td>8</td><td>90</td><td>60</td></tr><tr><td>Cold Stream 1</td><td>2.5</td><td>20</td><td>125</td></tr><tr><td>Cold Stream 2</td><td>3</td><td>25</td><td>100</td></tr></table>						Streams	Specific Heat, Cp	Source Temperature Ts (°C)	Target Temperature Tt (°C)	Hot Stream 1	2	150	60	Hot Stream 2	8	90	60	Cold Stream 1	2.5	20	125	Cold Stream 2	3	25	100
Streams	Specific Heat, Cp	Source Temperature Ts (°C)	Target Temperature Tt (°C)																						
Hot Stream 1	2	150	60																						
Hot Stream 2	8	90	60																						
Cold Stream 1	2.5	20	125																						
Cold Stream 2	3	25	100																						
VIII.	(a) Describe environmental auditing. Explain its significance.	4	L2	2	7																				
	(b) Explain air pollution, pollutants and their impacts.	5	L2	4	7,10																				
	(c) What are the sources of SO _x , NO _x and CO emissions?	3	L2	4	7,10																				
OR																									
IX.	(a) Give a brief description about different sources of emissions that pollute the environment.	4	L2	4	7,10																				
	(b) Explain the detailed procedure of exhaust emission test in a vehicle.	8	L2	4	7,10																				

Bloom's Taxonomy Levels

L1 = 13%, L2 = 50%, L3 = 11%, L4 = 18%, L5 = 8%.

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B.Tech. Degree VI Semester Regular Examination April 2022

ME 19-205-0607 ENERGY CONSERVATION AND ENVIRONMENT PROTECTION
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Gain a basic understanding and appreciation of energy efficiency and environmental concepts.

CO2: Learn basic operating principles of day to day energy conservation devices.

CO3: Discuss various options to increase energy efficiency.

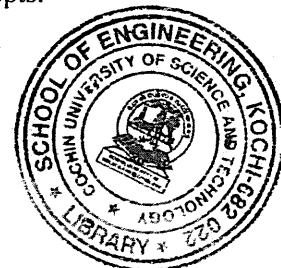
CO4: Examine ways to save energy and money.

CO5: Explore ways to save the environment.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome



PART A

(Answer *ALL* questions)

I.	(8 × 3 = 24)	Marks	BL	CO	PO
(a) List the strategies for better energy security of the nation.	3	3	L2	2	1,2
(b) List down the advantages of 'simple payback period' technique.	3	3	L2	1	1,2
(c) What is energy auditing? Discuss its general scope.	3	3	L1	1	1
(d) Enumerate the methods of testing steam traps.	3	3	L2	2	1,2
(e) Write short note on the environmental effects of nuclear power generation.	3	3	L1	1	1,2
(f) What are the implications of Global warming?	3	3	L2	1	1,2
(g) Explain the term Air Quality Index.	3	3	L1	1	1
(h) Discuss the impacts of water pollution.	3	3	L2	2	1,2

PART B

(4 × 12 = 48)

- II. (a) Give a detailed account of fossil fuel usage pattern among the world countries and give the limitations of fossil fuel. 4 L4 3 2,3
- (b) Calculate the NPV of two projects and suggest which of the two projects should be accepted assuming discount rate of 10%. 8 L5 4 3,4

	Project A	Project B
Initial Investment (₹)	40,000	60,000
Estimated life	5 years	5 years
Scrap Value (₹)	2,000	4,000

Cash inflows					
Year	1	2	3	4	5
Project A	10,000	20,000	20,000	6,000	4,000
Project B	40,000	20,000	10,000	6,000	4,000

OR

(P.T.O)

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- III. (a) Explain the trends in energy use patterns. 4 L4 3 1,2,3
- (b) Calculate the Internal Rate of Return of an investment of ₹1,36,000 which yields the following cash inflows. 8 L5 4 3,4

Year	Cash Inflows (₹)
1	30,000
2	40,000
3	60,000
4	30,000
5	20,000

- IV. Explain the ten steps methodology for the detailed energy auditing. 12 L4 4 3,4
- OR**
- V. (a) Distinguish between 'preliminary energy audit' and 'detailed energy audit'. 6 L4 3 3,4
- (b) Explain condensate recovery system in a steam system. 6 L4 4 3,4
- VI. Write the cogeneration systems based on: 12 L3 3 3,4
- (i) Diesel Engine
- (ii) Gas Turbine.
- OR**
- VII. Write short notes on: 12 L3 4 3,4
- (i) Geothermal energy
- (ii) Tidal energy.
- VIII. (a) Distinguish between primary and secondary air pollutants. 4 L4 5 4,5
- (b) What is the scope of environmental auditing in a country like India and explain the challenges? 8 L5 4 4,5
- OR**
- IX. (a) Explain the types of air pollutants and their impacts. 4 L4 5 4,5
- (b) Explain the detailed procedure of exhaust emission test in a vehicle. 8 L3 5 5

Blooms's Taxonomy Levels

L1-10%, L2-12.5%, L3-26.6%, L4-30%, L5-20%, L6-0%

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B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024

ME 19-205-0607 – ENERGY CONSERVATION AND ENVIRONMENT PROTECTION

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the overview of World Energy Scenario and energy economics.
 CO2: Study the Importance of energy management and energy auditing.
 CO3: Learn about the renewable energy sources.
 CO4: Understand the Environmental Impacts of energy use.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Classify Energy sources.	3	3	L1	1	1,3,7,10
(b) Define simple payback period.	3	3	L2	1	1,3,7,10
(c) Explain the need for energy audit.	3	3	L1	2	1,6,7,8,9
(d) Write any three energy conservation opportunity in pumping system.	3	3	L1	2	1,6,7,8,9
(e) Write the criteria for selection of wind mill installation.	3	3	L1	3	1,3,6,7, 11,12
(f) Explain principles of cogeneration.	3	3	L1	3	1,3,6,7, 11,12
(g) Define the term Environmental Auditing.	3	3	L1	4	1,7,8,10, 11
(h) What are the major Environmental effects of building Hydro Power Plants?	3	3	L1	4	1,7,8,10, 11

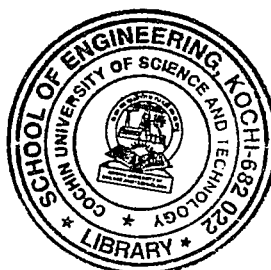
PART B

(4 × 12 = 48)

- II. Explain about Global Primary energy reserves. 12 L2 1 1,3,7,10

OR

(P.T.O.)



BTS-VI(R/S)-05-24-2481

	Marks	BL	CO	PO
III. Using NPV analysis technique, evaluate the financial merits of the two proposed projects given. Annual discount rate is 8% for both the projects.	12	L3	1	1,3,7,10

	Project 1	Project 2
Capital cost (₹)	30000	30000
Year	Net annual savings (₹)	Net annual savings (₹)
1	+6000	+6600
2	+6000	+6600
3	+6000	+6300
4	+6000	+6300
5	+6000	+6000
6	+6000	+6000
7	+6000	+5700
8	+6000	+5700
9	+6000	+5400
10	+6000	+5400
Total net savings at end of 10 th year	+60000	+60000

IV. Explain the steps and methodology for conducting the detailed Energy Audit.	12	L2	2	1,6,7,8,9
OR				
V. (a) List energy efficiency opportunities for refrigeration and air conditioning systems.	6	L2	2	1,6,7,8,9
(b) List various heat recovery devices and explain the working of Heat Wheels with diagram.	6	L2	2	1,6,7,8,9
VI. (a) With a diagram explain IC engine co-generation system.	6	L2	3	1,3,6,7,11,12
(b) Explain pinch, target setting and composite curves of heat exchangers.	6	L2	3	1,3,6,7,11,12
OR				
VII. Explain working of various types of Geothermal power plants with diagrams.	12	L2	3	1,3,6,7,11,12
VIII. (a) Explain the necessity of testing engine of vehicles and write the procedures for the test.	6	L2	4	1,7,8,10,11
(b) How do you describe the term emission factor and its significance with the help of an example?	6	L2	4	1,7,8,10,11
OR				
IX. How the energy usage deteriorates the environment? Explain with air and water pollution.	12	L2	4	1,7,8,10,11

Blooms's Taxonomy Level's

L1 – 17.5%, L2 – 72.5%, L3 – 10%.
